

SUMMARY

Software Development Engineer with 4+ years of experience building Java/Spring Boot microservices for fintech and enterprise platforms. Design and ship event-driven, cloud-native systems on AWS and Azure that handle high-volume payment processing, improve fraud detection, and ensure scalability with Docker, Kubernetes, and CI/CD pipelines, including AI-powered voice agents and multi-agent applications for real-time and conversational use cases.

EDUCATION

Master Of Computer Science | George Mason University

05/2025

SKILLS

Programming Languages: Java, Python, C, C++, JavaScript (ES6+), TypeScript, SQL

Backend Development: Spring Boot, FastAPI, Node.js, Express.js, RESTful APIs, Microservices, Apache Kafka

Frontend Development: React.js, Angular, HTML5, CSS3, Bootstrap

Cloud/DevOps Technologies: AWS (EC2, S3, RDS, Lambda, EKS), Jenkins, GitHub Actions, Docker, Maven

Database & Caching: PostgreSQL, MySQL, MongoDB Oracle, Redis, Query Optimization, Data Modeling

Monitoring & Logging: Prometheus, Grafana, ELK Stack

Testing/Quality: JUnit, Postman, Swagger

AI/ML & Data Engineering: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Machine Learning Basics, LLM Integration, RAG, AI-Assisted Development

Tools & Process: Git/GitHub, Jira, Agile/Scrum, SDLC

Certifications: AWS Certified Solution Architect – Associate, AWS Certified Cloud Practitioner

EXPERIENCE

Software Development Engineer | PNC, USA

07/2025 – Present

- Engineered cloud-native microservices using Java, Spring Boot, and Python (FastAPI) to process real-time financial transactions, to improve fraud detection pipelines, to enhance decision accuracy for risk analytics teams handling millions of daily transactions.
- Designed and implemented event-driven architecture using Apache Kafka and AWS Lambda, to enable real-time streaming of payment and account data, to reduce processing latency by 35% and improve responsiveness for internal fraud monitoring systems.
- Developed AI-assisted fraud detection services leveraging LLMs and RAG pipelines with Python and vector embeddings, to analyze transactional anomalies, to increase fraud detection precision by 20% for banking operations teams.
- Built and deployed scalable APIs using Node.js, Express, and Spring Boot, to integrate third-party payment gateways and internal banking services, to ensure seamless transaction orchestration for customer-facing platforms.
- Containerized and orchestrated distributed services using Docker and Kubernetes (EKS), to improve deployment consistency and scalability, to reduce infrastructure provisioning time by 40% for DevOps and platform engineering teams.
- Automated CI/CD pipelines using Jenkins and GitHub Actions, to streamline build, test, and deployment workflows, to reduce release cycles by 30% and improve delivery efficiency for cross-functional engineering teams.
- Implemented centralized monitoring and observability using Prometheus, Grafana, and ELK Stack, to track system health and logs, to reduce MTTR by 25% for production support teams.

Software Engineer | Razorpay, India

08/2021 – 07/2023

- Developed scalable backend services using Java, Spring Boot, and Hibernate, to handle high-volume payment processing systems, to support millions of transactions per day for merchants and financial partners.
- Architected RESTful APIs and microservices using Node.js and Express, to enable seamless integration with payment gateways and banking APIs, to enhance transaction success rates and user experience for end customers.
- Implemented event-driven systems using Apache Kafka, to process real-time payment events and webhook notifications, to improve system throughput by 30% for payment processing pipelines.
- Optimized database performance using PostgreSQL and Redis caching, to reduce query latency and improve API response times by 40%, to enhance platform performance for merchant dashboards.
- Built responsive frontend components using React.js and Angular, to create real-time payment analytics dashboards, to enable faster decision-making for business and operations teams.
- Containerized services using Docker and deployed on AWS (EC2, S3, RDS), to ensure high availability and scalability, to support growing transaction volumes across regions.
- Established CI/CD pipelines using Jenkins, to automate testing and deployments, to reduce manual intervention and improve release reliability for engineering teams.

PROJECTS

Clip Lens

[GitHub](#)

- Built a multi-agent AI web app that transforms any YouTube lecture URL into a structured study environment, orchestrating 5 specialized agents (transcript fetch, summarization, semantic search, translation, faculty audit) over a FastAPI SSE streaming backend.
- Designed a provider-agnostic LLM layer supporting OpenAI GPT-4o, with ChromaDB in-memory vector indexing for semantic search and even-sampling logic to handle long videos within context window limits.
- Deployed to Railway via GitHub CI/CD with automatic redeloys, implementing stateless privacy-by-design (no database, no user accounts) and a dual Student/Faculty mode with real-time agent progress animations, flashcard generation, and PDF audit reports.

Voice Bot

[GitHub](#)

- Built a low-latency, browser-based voice assistant with a modular STT → LLM → TTS pipeline at 16kHz, using AudioWorklet and WebSockets to enable real-time streaming conversations with live transcription and barge-in interruption support.
- Designed a provider-agnostic Python aiohttp backend that decouples speech and language layers, supporting Deepgram/Azure Speech for STT/TTS and OpenAI GPT-4/Azure OpenAI for inference, with full configurability via environment variables and swappable system prompts.
- Containerized the service with Docker and deployed to Azure Container Apps on a single port (8080) for HTTP and WebSocket traffic, with connection lifecycle management, graceful error handling, optional audio recording, and health checks for production reliability.